

TEAM NAME: ELITE CODERS

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PROJECT TITLE: JET CULTIVATION

EXPERIMENT: 2: PROBLEM STATEMENT, OBJECTIVE, FUNCTIONAL AND NON - FUNCTIONAL REQUIREMENTS, USER STORY AND ACCEPTANCE CRITERIA

Problem Statement:

Farmers in Tamil Nadu often face difficulties in making informed agricultural decisions due to a lack of integrated, location-specific information. Many farmers are unaware of nearby water sources such as rivers, the current status of those rivers, suitable irrigation methods for their soil type, and the crops best suited for their land. Additionally, guidance from seed planting to harvesting is often fragmented or unavailable. Even after successful cultivation, farmers struggle to identify nearby and optimal centers to sell their produce at fair prices. This lack of a unified digital platform results in inefficient water usage, improper crop selection, lower yields, and reduced income for farmers.

Objective:

The main objective of the Farm Flow application is to provide a smart, location-based agricultural support system for farmers in Tamil Nadu. The app aims to assist farmers in:

- * Selecting suitable irrigation methods based on river status and soil type
- * Identifying appropriate crops for their land
- * Receiving end-to-end guidance from planting to harvesting
- * Finding optimized locations to sell their harvested crops

Overall, the goal is to improve agricultural productivity, optimize resource usage, and enhance farmers' income through informed decision-making.

Functional requirements:

1. User Location Selection

The system shall allow users to select their district within Tamil Nadu.

2. River Identification The system shall identify and display nearby rivers based on the selected district.

3. River Status Monitoring

The system shall analyze and display the current status of the selected river.

4. Irrigation Recommendation

The system shall recommend suitable irrigation methods based on river status and soil type.

5. Soil Type Analysis

The system shall determine and display the soil types present in the selected land area.

6. Crop Recommendation

The system shall suggest suitable crops based on soil type and regional conditions.

7. Crop Selection

The system lets users choose a crop from the recommended list.

8. Crop Lifecycle Guidance

The system provides step-by-step guidance from seed selection and planting to harvesting.

9. Farming Alerts and Tips

The system gives timely tips related to irrigation, fertilizers, and pest control.

10. Market Identification

The system identifies nearby markets or selling centers for harvested crops.

Non-functional requirements :

1. Usability

The application must have a simple, user-friendly interface that farmers with minimal technical knowledge can easily understand and operate.

2. Performance

The system shall process requests (river status checking, soil analysis, crop recommendations) within a reasonable response time.

3. Security

The application shall ensure secure handling of user data (location & farming details) through authentication and protected storage.

4. Reliability

The system shall operate consistently and provide accurate information even with unstable or limited network connectivity.

5. Scalability

The application shall be capable of handling increased load and data volume without affecting performance.

Farmer Login & Registration

User Story

As a farmer, I want to register and log in so that I can securely access the system.

Acceptance Criteria

User can register with valid details

User can log in with correct credentials

Invalid login shows error message

Successful login redirects to dashboard

Story Points

3

Priority

4

Risk

2 – Medium

Dashboard Navigation:

User Story

As a farmer, I want to view a dashboard so that I can access different features easily.

Acceptance Criteria

Dashboard loads after login

Cards are visible:

- Soil & Crops
- River & Irrigation
- Crop Journey
- Selling Centers

Each card is clickable.

Story Points

2

Priority

3

Risk

3 – Low

Soil Data & Suitable Crops

User Story

As a farmer, I want to view soil data and suitable crops so that I can improve yield.

Acceptance Criteria

Soil details are displayed (type, moisture)

Suitable crops are shown

Data is readable and structured

Page loads without errors.

Story Points

5

Priority

2

Risk

1 – High

River Monitoring Status

User Story

As a farmer, I want to check river status so that I can plan irrigation.

Acceptance Criteria

River status is displayed

Data is updated/simulated correctly

Page loads properly.

Story Points

3

Priority

3

Risk

1 – High

Irrigation Suggestions

User Story

As a farmer, I want to check river status so that I can plan irrigation.

Acceptance Criteria

River status is displayed

Data is updated/simulated correctly

Page loads properly.

Story Points

3

Priority

3

Risk

1 – High

Crop Journey Tracking

User Story

As a farmer, I want to view crop journey stages so that I can manage cultivation effectively.

Acceptance Criteria

Crop stages are displayed

Each stage has description

Navigation works properly.

Story Points

3

Priority

2

Risk

3 – Low

User Story

As a farmer, I want to find optimal selling centers so that I can maximize profit.

Acceptance Criteria

Selling centers are listed

Location details are shown

Data is clearly displayed

Story Points

3

Priority

2

Risk

2 – Medium

EliteCoders2512 / JetCultivation / Boards / Backlogs

JetCultivation

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

Artifacts

Project settings

JetCultivation Team

View as Board
Column Options

Backlog
Analytics

+

-

Order

Work Item Type

Title

State

Effort

Busin...

Value

1	Epic	Smart Jet Cultivation Management System	New	Busin...
	User Story	Farmer Login & Registration	New	Busin...
	User Story	Dashboard Navigation	New	Busin...
	User Story	Soil Data & Suitable Crops	New	Busin...
	User Story	River Status Monitoring	New	Busin...
	User Story	Irrigation Suggestions	New	Busin...
	User Story	Crop Journey Tracking	New	Busin...
	User Story	Optimal Selling Centers	New	Busin...

Epics

Iterations

Sprints

Planning

Drag and drop work items to include them in a sprint.

JetCultivation Team backlog

Iteration 1

No work scheduled yet

Iteration 2

No work scheduled yet

Iteration 3

No work scheduled yet

+ New Sprint

USER STORY 4

4 Farmer Login & Registration

No one selected

0 Comments

Add Tag

Save and Close

Follow

Refresh

Share

More

State: New

Area: JetCultivation

Reason: New

Iteration: JetCultivation

Updated by Monisha S: Mar 21

Details

1

0

Description

As a farmer, I want to register and log in so that I can securely access the system.

Acceptance Criteria

User can register with valid details
 User can log in with correct credentials
 Invalid login shows error message
 Successful login redirects to dashboard

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

[switch to Markdown editor](#)

Planning

Story Points: 3
 Priority: 4
 Risk: 2 - Medium

Classification

Value area: Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Parent
 Smart Jet Cultivation Management System
 Updated Mar 23 • New